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Batch:CH3

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

m is the mass of the object (in kilograms).

v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 # Accept mass and velocity as input
2 mass = float(input())
3 velocity = float(input())
4
5 # Calculate momentum
6 momentum = mass * velocity
7
8 # Display result rounded to 2 decimal places with unit
9 print(f"{momentum:.2f}kgm/s")
10
```

Average time

0.002 s

2.25 ms

Maximum time

0.003 s

3.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

5.8

10.8

50.00kg/s

Actual output

5.8

10.8

50.00kg/s

Test case 2

Terminal

Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

$1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2
3 # Accept input
4 num = int(input())
5
6 # Check number of digits and perform operation
7 if 1 <= num <= 9:
8     # Single-digit -> square
9     print(num ** 2)
10
11 elif 10 <= num <= 99:
12     # Two-digit -> square root (2 decimal places)
13     print(f"{math.sqrt(num):.2f}")
14
15 elif 100 <= num <= 999:
16     # Three-digit -> cube root (2 decimal places)
17     print(f"{num ** (1/3):.2f}")
18
19 else:
20     print("Invalid")
```

Average time

0.002 s

2.14 ms

Maximum time

0.004 s

4.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

Expected output

81

Actual output

81

Test case 2

Test case 3

Terminal

Test cases

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 from datetime import datetime
2
3 # Read input dates
4 date1 = input().strip()
5 date2 = input().strip()
6
7 # Convert string to datetime object
8 d1 = datetime.strptime(date1, "%Y-%m-%d")
9 d2 = datetime.strptime(date2, "%Y-%m-%d")
10
11 # Calculate difference
12 difference = d2 - d1
13
14 # Print number of days
15 print(difference.days)
16
17 from datetime import date
```

Average time

0.007 s

7.00 ms

Maximum time

0.019 s

19.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 2

Terminal

Test cases

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 # Read input
2 num = int(input().strip())
3
4 rev = 0
5
6 # Reverse using arithmetic
7 while num > 0:
8     digit = num % 10
9     rev = rev * 10 + digit
10    num = num // 10
11
12 print(rev)
13
```

Average time0.002 s2.00 msMaximum time0.003 s3.00 ms

2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1

Expected output53677635Actual output53677635

Test case 2

TerminalTest cases

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:
<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 num = int(input())
2
3
4 # Print multiplication table from 1 to 10
5 for i in range(1, 11):
6     print(f"{num} x {i} = {num * i}") # Take integer input
7
8
9
```

Average time0.002 s1.75 msMaximum time0.002 s2.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1

Expected output88 x 1 = 88 x 2 = 168 x 3 = 248 x 4 = 328 x 5 = 40Actual output88 x 1 = 88 x 2 = 168 x 3 = 248 x 4 = 328 x 5 = 40

TerminalTest cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

- If the student passes all courses:
 - Print the aggregate percentage (formatted to two decimal places).
 - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
 - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

```
1 # Read number of courses
2 n = int(input().strip())
3
4 # Read marks
5 marks = list(map(int, input().strip().split()))
6
7 # Check if any subject is failed
8 if any(mark < 40 for mark in marks):
9     print("Fail")
10 else:
11     # Calculate aggregate percentage
12     aggregate = sum(marks) / n
13
14     # Print aggregate percentage
15     print(f"Aggregate Percentage: {aggregate:.2f}")
16
17     # Determine grade
18     if aggregate > 75:
19         grade = "Distinction"
20     elif 60 <= aggregate < 75:
21         grade = "First Division"
22     elif 50 <= aggregate < 60:
23         grade = "Second Division"
24     elif 40 <= aggregate < 50:
25         grade = "Third Division"
26     else:
27         grade = "Fail"
```

Average time0.003 s2.80 msMaximum time0.004 s4.80 ms

5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed

Test case 1

Expected output455 65 78 89Aggregate Percentage: 71.75Grade: First DivisionActual output455 65 78 89Aggregate Percentage: 71.75Grade: First Division

Test case 2

TerminalTest cases

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

A single integer n representing the number of terms to generate.

Output Format:

A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n == 0:
3         return 0
4     elif n == 1:
5         return 1
6     else:
7         return fibonacci((n - 1) + fibonacci(n - 2))
8
9
10
11 n = int(input())
12 for i in range(1, n + 1):
13     print(fibonacci(i), end=" ")
14
15 def fibonacci(n):
16     if n == 0:
17         return 0
18     elif n == 1:
19         return 1
```

Average time

0.006 s

6.25 ms

Maximum time

0.017 s

17.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

0 1 1 2 3

Actual output

0 1 1 2 3

Test case 2

Terminal

Test cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 # Read number of rows
2 n = int(input().strip())
3
4 # Print right-angled triangle pattern
5 for i in range(1, n + 1):
6     for j in range(i):
7         print("*", end=" ")
8     print()
```

Average time

0.002 s

1.67 ms

Maximum time

0.003 s

3.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1

Expected output

*
* *
* * *
* * * *

Actual output

*
* *
* * *
* * * *

Terminal

Test cases

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 # Read number of rows
2 n = int(input().strip())
3
4 # Print pattern
5 for i in range(1, n + 1):
6     for j in range(1, i + 1):
7         print(j, end=" ")
8     print()
```

Average time

0.003 s

2.75 ms

Maximum time

0.006 s

6.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

Actual output

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

Terminal

Test cases